

UNIVERSITY OF CAMBRIDGE

NATURAL SCIENCES TRIPOS Part II

Wednesday 10th June 2026

09:00 to 11:00

PHYSICS

PHYSICAL SCIENCES: HALF SUBJECT PHYSICS

ADVANCED AND QUANTUM OPTICS — Paper 4

*Candidates offering this paper should attempt **all six** questions (four from Section A and two from Section B).*

The approximate number of marks allocated to each question or part of a question is indicated in the right margin. This paper contains 4 sides, including this coversheet, and is accompanied by a handbook giving values of constants and containing mathematical formulae which you may quote without proof.

STATIONERY REQUIREMENTS

2 × 8-page answer booklet

Rough workpad

Yellow master coversheet

SPECIAL REQUIREMENTS

Mathematical Formulae Handbook

Approved calculator allowed

You may not start to read the questions printed on the subsequent pages of this question paper until instructed that you may do so by the Invigilator.

SECTION A

Attempt **all four** questions in this Section. Answers should be concise and relevant formulae may be assumed without proof.

1 A circular aperture of diameter $D = 0.50$ mm is illuminated by a uniform monochromatic plane wave of wavelength $\lambda = 633$ nm. Calculate the Fresnel number N_F at distances $d_1 = 5$ cm and $d_2 = 5$ m, and state in which regime (Fresnel or Fraunhofer) each lies. For the far-field case, calculate the radius of the central Airy disk on a screen at that distance.

2 A two-level atom in a cavity of volume V has a transition cross-section $\sigma(\nu_0) = 1.5 \times 10^{-17}$ m² at line centre and a spontaneous-emission lifetime $t_{\text{sp}} = 25$ ns. Using the standard relation

$$\sigma(\nu) = \frac{\lambda^2}{8\pi t_{\text{sp}}} g(\nu),$$

estimate the linewidth $\Delta\nu$ of the transition at $\lambda_0 = 780$ nm, assuming a Lorentzian lineshape for which $g(\nu_0) = 2/(\pi\Delta\nu)$. Comment on whether your result is consistent with the natural linewidth $\Delta\nu_{\text{nat}} = 1/(2\pi t_{\text{sp}})$.

3 An object transparency consists of a one-dimensional sinusoidal amplitude grating with transmittance $t(x) = \frac{1}{2}[1 + \cos(2\pi x/\Lambda)]$ and period $\Lambda = 20$ μm . It is placed in the front focal plane of a thin lens of focal length $f = 200$ mm and illuminated by a normally incident plane wave at $\lambda = 500$ nm. Determine the positions and relative intensities of the bright spots that appear in the back focal plane of the lens.

4 A confocal optical resonator consists of two identical concave mirrors of radius of curvature $R = 50$ cm placed exactly $d = R$ apart. Calculate the values of the g -parameters and verify that the resonator lies on the boundary of the stability region. For a Gaussian beam mode of wavelength $\lambda = 1064$ nm in this resonator, calculate the waist radius W_0 of the fundamental mode (located at the centre of the resonator).

SECTION B

Attempt **both** questions from this Section.

5 Spatial filtering in a $4f$ optical processor.

A coherent optical processor consists of two identical lenses of focal length f separated by $2f$. An object transparency with amplitude transmittance $f(x, y)$ is placed in the front focal plane of the first lens, a filter mask $H(\nu_x, \nu_y)$ is placed in the common (Fourier) plane between the lenses, and the output is observed in the back focal plane of the second lens. The system is illuminated by a uniform monochromatic plane wave of wavelength λ .

- (a) State, in your own words, the Fourier-transform property of a thin lens. Hence show that, in the absence of the filter mask, the field at the output plane is a (suitably scaled) inverted image of $f(x, y)$ — i.e. the system acts as a unit-magnification imaging system.
- (b) The filter mask in the Fourier plane is a circular opaque stop of radius r_0 , centred on the optical axis (a high-pass filter that blocks the low spatial frequencies). Show that this filter blocks all spatial frequencies satisfying

$$\sqrt{\nu_x^2 + \nu_y^2} < \frac{r_0}{\lambda f}.$$

Sketch qualitatively what happens to the image of a circular aperture of diameter $D \gg \lambda f/r_0$ when this filter is applied, explaining the physical origin of the edge enhancement effect.

- (c) The object is replaced by a periodic line grating with transmittance $f(x, y) = \frac{1}{2}[1 + \cos(2\pi x/\Lambda)]$, illuminated normally. By computing the Fourier transform of $f(x, y)$, determine the locations of the spots that appear in the Fourier plane. The opaque stop of part (b) now blocks only the central spot ($\nu_x = \nu_y = 0$). Show that the resulting output intensity is uniform, and find its value in terms of the input intensity I_0 .
- (d) Now suppose the filter is instead a small *transparent* hole at the position ($\nu_x = 1/\Lambda$, $\nu_y = 0$) in the Fourier plane, blocking all other components. Determine the resulting field amplitude at the output plane. Comment on what this demonstrates about how spatial filtering can extract specific spatial-frequency content from an image.

6 Holography of a point source and image reconstruction.

A hologram is recorded by interfering an object wave $U_o(x, y)$ at a photographic plate (in the plane $z = 0$) with a reference plane wave

$$U_r(x, y) = A_r \exp(-jk \sin \theta x),$$

which is incident on the plate at an angle θ in the xz -plane. The plate records an intensity pattern proportional to $|U_o + U_r|^2$, and after development its amplitude transmittance is taken to be linearly proportional to the recorded intensity:

$$t(x, y) = t_0 + \beta |U_o(x, y) + U_r(x, y)|^2,$$

where t_0 and β are constants.

- (a) Expand $|U_o + U_r|^2$ and hence show that the transmittance $t(x, y)$ contains four distinct terms. Identify which term carries the information needed to reconstruct the original object wave, and which term is responsible for the so-called *conjugate* (or *twin*) image.
- (b) The hologram is now illuminated for reconstruction by an exact replica of the original reference wave U_r . Show that the transmitted field at the hologram plane consists of four contributions, and identify which one reproduces the original object wave (multiplied by a constant). Explain how the angle θ of the reference wave allows the reconstructed object wave to be physically separated from the conjugate image and from the directly transmitted reference beam, in the spirit of off-axis (Leith–Upatnieks) holography.
- (c) Now suppose the object is a single point source located at $(x, y, z) = (0, 0, -z_0)$, where $z_0 > 0$. Within the Fresnel approximation, the wave it produces at the hologram plane is

$$U_o(x, y) = \frac{A_o}{z_0} \exp(jkz_0) \exp\left[\frac{jk(x^2 + y^2)}{2z_0}\right].$$

Substitute this into the transmittance expression and show that the term containing $U_r^* U_o$ takes the form of a parabolic phase factor multiplied by an off-axis linear phase. Identify the focal length and lateral position of the lens-like phase pattern recorded on the hologram.

- (d) When this hologram is illuminated with the reference wave, the reconstructed field forms a real image of the original point source. By regarding the hologram as an off-axis Fresnel zone plate (acting as a thin lens of the focal length you found in part (c)), state where the real and virtual images are located relative to the hologram plane, and explain why an in-line (Gabor) hologram suffers from the unwanted superposition of the object, conjugate, and reference beams whereas the off-axis arrangement of part (b) does not.

END OF PAPER